

# **ICT702 Introduction to the Relational Database Assessment 2 T3-25**

## **Assignment 2-Individual Assignment –Database Design and Implementation**

### **Purpose and Learning Outcomes**

#### **Purpose**

The purpose of the assignment is to provide students with the opportunity to apply knowledge and skills developed during the semester with reference to:

1. Develop a Case Study
2. Interpretation of business rules from a case study.
3. Conceptual data modelling through the creation of an Entity Relationship (ER) model.

### **Timelines and Expectation**

The percentage value of this Assignment is 10% of the course mark.

Due: the due date is on Sunday, 7/12/2025 23:59, week 6. Refer to the course outline.

Minimum Time Expectation: 15-20 hours.

### **Case Study Requirements**

1. Scenario choice: your chosen scenario must include at least Ten to Twelve entities that are related to each other. This can be anything you choose, but you are advised to choose something you know, to make the exercise easier. An example could be a hobby you have – perhaps you enjoy a sport, and you wish to develop a database that models some form of performance related to the teams and games played in this sport. Another example could be a part-time job, and you may wish to develop a database to assist some aspect of that business.

Another example could be some area of science, perhaps an engineering-type application that

models a power station or a factory or an experiment of some kind. Because this “brief” is very open-ended, you are recommended to check your chosen scenario with the teaching staff to make sure it is suitable. A final consideration when thinking of your scenario is the list of SQL queries that you will need to execute within your database to prove the creation of tables and insertion of data. For instance, you will need to include certain search and statistical functions, so your scenario will need to be sophisticated enough to make these queries possible. PLEASE NOTE: each student is required to develop a UNIQUE scenario – it will not be permitted for two students to use the same scenario.

2. Your ER Diagram must include **Ten to Twelve entities**. There is no upper bound on how many entities you choose, but you are advised to include no more than **Fifteen**. Design your ER Diagram with all entity names, attribute names, primary and foreign keys, relationships, cardinality, and participation indicated. You will need to normalize all your entities, to resolve any many-to-many relationships.

Observe the following restrictions when creating your scenario:

- Include a specialization hierarchy, with super types and overlapping or disjoint sub-types.
- Include an example of both composite and surrogate primary keys.
- One (or more) of your entities must have a numeric field.
- One (or more) of your entities must contain an alphanumeric (varchar) field.
- One (or more) of your entities must contain a Date field.
- Your attribute names and primary and foreign keys should be indicated as per the conventions given in the lecture slides (i.e. attributes as proper nouns, **primary key underlined**, and foreign keys in italics).

## Assignment Requirements

### Detailed Requirements

For the Draft Assessment, Students are expected to develop and provide their own case study, and then provide the following:

- 1- Case Study
- 2- Entity Name
- 3- Attributes
- 4- Entity Relations
- 5- Weak and Strong Entity
- 6- Cardinalities and Connectivity
- 7- Optional and Mandatory Relationships
- 8- Primary and Foreign Key

| <b>Marking Guide</b> |                                                                                                                                                                                                                                                                                                                                                                                                       |           |              |
|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|--------------|
| Requirements         | Description                                                                                                                                                                                                                                                                                                                                                                                           | Max Marks | Student Mark |
|                      | <ul style="list-style-type: none"> <li>• Proper and Valid Case Study (10-12) Entities, including supertype and Subtype, Overlap or disjoint.</li> <li>• Entities Name</li> <li>• Attributes</li> <li>• Primary and Foreign Key's</li> <li>• Entity Relations</li> <li>• Weak and Strong Entities</li> <li>• Cardinalities and Connectivity</li> <li>• Optional and Mandatory Relationships</li> </ul> | 10        |              |
| Total Mark           |                                                                                                                                                                                                                                                                                                                                                                                                       | 10        |              |
| Course Mark          |                                                                                                                                                                                                                                                                                                                                                                                                       | 10        |              |